

Pre-Requisites

To effectively understand this session, learners are expected to have:

- Python programming knowledge (functions, OOP, APIs, JSONs)
- Familiarity with Large Language Models (prompting, responses)
- Prior exposure to utilising LLMs from LangChain, Google Gemini SDK
- Exposure to basic tool-calling in LLMs

Session Summary

This 180-minute live session introduces learners to the foundations of agentic systems, focusing on how agents differ from traditional pipelines and how they operate through iterative reasoning and action loops.

The session begins by establishing a clear conceptual understanding of agentic behaviour through comparative analysis. Learners then construct an agent from first principles, with particular emphasis on state management as the core of agent intelligence.

Building on this, the session explores tool integration within the agent loop, including schema design and validation across different LLM interfaces. Learners then progress to agent skills as composable abstractions, followed by control mechanisms such as termination conditions, trace logging, and reflection. The session concludes with a structured analysis of common failure modes and debugging strategies.

Learning Objectives/Outcomes

By the end of this session, learners will be able to:

- Define and identify agentic systems based on key properties such as goal-directed behaviour, state, and iterative reasoning
- Construct an agent loop from scratch, including explicit state representation and updates
- Integrate tools and skills into the agent loop and design robust tool schemas with validation mechanisms
- Implement control mechanisms including termination conditions, trace logging, and reflection loops

Brief Agenda

Learning Objective	Comments	Duration	Mode
Understand agentic behaviour	Identify agentic vs non-agentic systems	20 mins	Conceptual + Guided Analysis
Build agent loop with state	Implement loop and manage state	30 mins	Demonstration + Guided Coding
Integrate tools & schemas	Add tools, define schemas, validate	40 mins	Guided Coding
Develop agent skills	Build and compose skills	40 mins	Demonstration + Guided Practice
Control & tracing	Implement termination, trace, reflection	25 mins	Demonstration
Analyse failures	Identify and debug agent failures	20 mins	Guided Analysis
Wrap-up	Summarise and resolve doubts	5 mins	Q&A

Conclusions

By the end of this live session, learners will understand that:

- The agent loop and state form the foundation of all agentic systems
- Tools extend capabilities, but structure (schemas + validation) ensures reliability
- Skills enable modular and scalable agent design
- Control mechanisms and trace are essential for debugging and robustness

Additional Readings

- [Agents - Docs by LangChain](#)
- [Tools - Docs by LangChain](#)
- [Agents Overview | Gemini API | Google AI for Developers](#)
- [Using Tools with Gemini API | Google AI for Developers](#)
- [Function Calling | Google Gen AI SDK documentation](#)
- [Agents SDK | OpenAI API](#)
- [Function calling | OpenAI API](#)
- [Using tools | OpenAI API](#)